Can a Quantum Circuit Read a Sentence?

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Turning "the clever program runs" into a trained 2-qubit circuit — worked end to end

Physics 14N · Stanford University

The goal

Read a short sentence and decide **positive** or **negative** (its **sentiment**) — but do it the way **Quantum NLP** does: turn the words into a **quantum circuit** and let a measurement give the answer.

positive=**good**=**YES**$=1$ | **negative**=**bad**=**NO**$=0$

The whole pipeline in one line

words → dials · grammar → wiring & weights · circuit → measure → answer

Each word owns one **learned number**. Grammar decides how those numbers are combined. A quantum measurement turns the result into a probability of "**positive**."

Step 1 — Grammar gives each word a role *and* a weight

Our sentences follow one shape; each word plays a colored **role**:

clever **program** **runs** **(the puzzle)**

adjective subject verb object

Grammar says *who connects to whom* — and **how much each role counts**. The verb and adjective carry sentiment; the bare nouns barely matter:

role	weight
verb (the action)	**1.40**
adjective (describes)	**1.25**
object-adjective	0.90
subject (the thing)	0.45
object (acted on)	0.35

(Object roles only kick in for longer sentences with an object — ours, "the clever program runs," uses just the top, subject, and verb.)

Step 2 — A qubit is a dial, and measuring squares it

A **bit** is a switch: **0** or **1**. A **qubit** is a **dial** that can point anywhere between **NO** (0) and **YES** (1). **What's quantum:** until you look it holds a real **blend** of both at once.

- A dial tilted by angle θ carries **amplitude** $\cos(\theta/2)$ on NO and $\sin(\theta/2)$ on YES.
- **Measuring is random; the rule is: probability = amplitude^2.** A dial leaning far to YES has a big YES-amplitude ⇒ reads YES almost every time. Straight up = 50/50.

We use **two** dials: a **subject dial** and a **sentence dial** (it ends up holding the answer). For both dials at once, **multiply** their amplitudes for each reading.

Step 3 — Turning words into rotation angles

Each word's learned number is how far it **rotates** a dial. Grammar adds the right words onto each wire. For **the clever program runs**, the *trained* dials are **clever**$=0.098$, **program**$=0.043$, **runs**$=0.489$:

$$\textbf{subject angle} = \text{clever} + \text{program} = 0.098 + 0.043 = \mathbf{0.141}$$

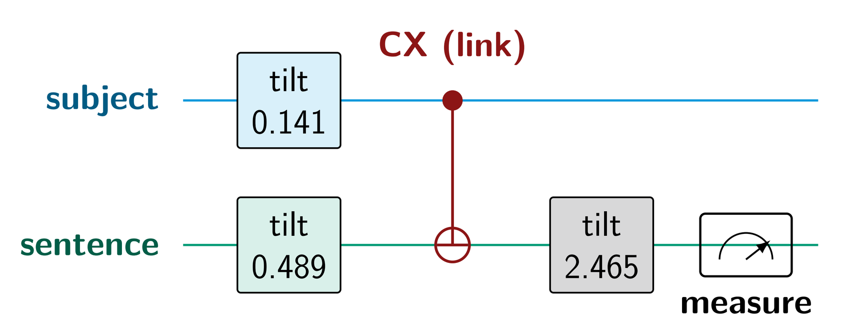
$$\textbf{sentence angle} = \text{runs} = \mathbf{0.489}$$

$$\textbf{decision angle} = \underbrace{\pi/2}_{\text{start straight up}} + \underbrace{\text{score}}_{\text{weighted words}}$$

$$\text{score} = \underbrace{0.068}_{\text{bias}} + 1.25(0.098) + 0.45(0.043) + 1.40(0.489) = 0.894$$

$$\text{decision angle} = \pi/2 + 0.894 = \mathbf{2.465}$$

The decision dial starts **straight up** (the 50/50 line) so the weighted words can push it either way — and the **verb** pushes hardest (weight 1.40).



"**tilt** θ" just means *rotate that dial by angle* θ (physicists call it an *Ry* gate). The sentence dial is tilted **twice**: first by its own word (0.489), then by the whole weighted verdict (2.465).

Step 4 — The CX gate: grammar's glue (and the quantum part)

Grammar says the **subject** and the **action** are **one** sentence, so we link their dials with a **CX gate**. Its rule (**C** = control/check, **X** = flip):

If the subject reads YES, flip the sentence dial; if NO, leave it.

Literally, on the four readings

The two dials together have four readings (left digit = subject, right = sentence). CX just **swaps the two where the subject reads 1** ($10 \leftrightarrow 11$):

	00	01	10	11
before	0.968	0.241	0.068	0.017
after	0.968	0.241	0.017	0.068

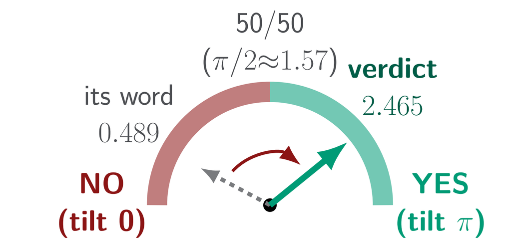
Because each dial is still a **blend**, this welds their answers together — like **two coins glued**: you can't flip one without the other. They are now **entangled**, and no separate spinning can fake that link.

Watch one sentence compute, slot by slot

The two dials together have four possible readings: **00, 01, 10, 11** (**left digit** = **subject dial**, **right digit** = **sentence dial**). The circuit tracks one **amplitude** per reading; each step just reshuffles these four numbers. Read the table as a sentence per row:

what we do	00	01	10	11
both dials start at NO	1.000	0.000	0.000	0.000
tilt *each* dial by its own words	0.968	0.241	0.068	0.017
CX links them: swap 10 & 11	0.968	0.241	0.017	0.068
tilt the sentence dial by the verdict	0.094	0.993	-0.059	0.039

A "tilt" is an **angle**: $0=$ NO, $\pi\,(\approx 3.14) =$ YES, and the halfway $\pi/2\,(\approx 1.57) =$ 50/50. The **sentence dial**'s own word barely tilts it (0.489 — still in NO); the **verdict** (2.465) clears the 50/50 line and lands it in **YES**:



Now measure the sentence dial. It reads **YES** in the readings whose right digit is 1 — the **green** columns **01** and **11**. Square those amplitudes and add:

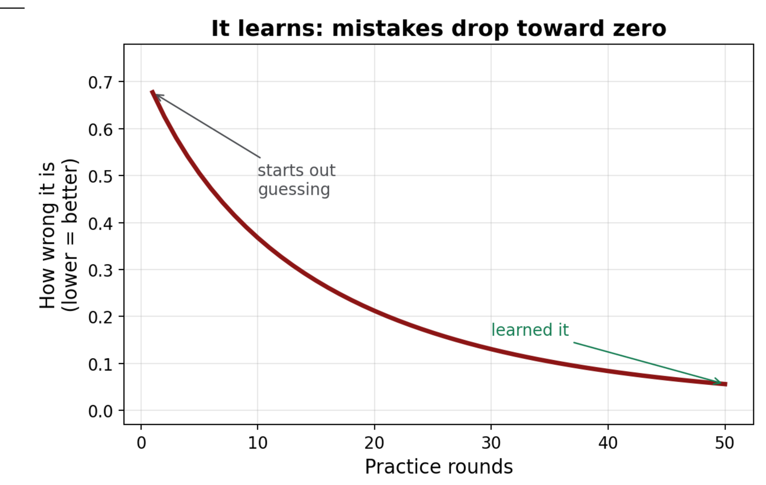
$$P(\text{YES}) = 0.993^2 + 0.039^2 = \mathbf{0.988} \;\Rightarrow\; \textbf{POSITIVE} \checkmark$$

Swap in **the broken code fails** and the same circuit lands at $P(\text{YES}) =$ **0.003** ⇒ **NEGATIVE**.

Does it work?

	Correct
Practice (60)	**100%**
New, unseen (12)	**100%**

Every answer comes from **simulating and measuring** this circuit. With only **2 qubits** a laptop does it exactly in <1 s; real language needs *many* linked qubits — where a true quantum computer becomes the point.



The update rule, explicitly

Run the circuit, get $P(\text{YES})$; the **miss** is $e = P(\text{YES}) - \text{label}$. For each dial's angle θ, get its **slope** by re-running the circuit with that angle shifted $\pm\frac{\pi}{2}$, then step the dial against the miss:

$$\text{slope} = \tfrac{1}{2}\left[\,P(\theta+\tfrac{\pi}{2}) - P(\theta-\tfrac{\pi}{2})\,\right]$$

$$\text{dial} \leftarrow \text{dial} - \eta\,(\,e \times \text{slope}\,)$$

The shift-and-subtract is the *parameter-shift rule* (how real quantum circuits get their gradients); η is a small learning rate. Repeat over every sentence for ~50 rounds.

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